(a) (b) (c) (d)

$H = (\Omega/2)\sum X + \sum U_{ij} n_i n_j$
$U_{ij} \propto [1 + (r_{ij}/R_c)^6]^{-1}$
$\Omega = 0.05\, U_{nn}$, from $|+x\rangle$

R6: R_c/a

time t $[1/V_{nn}]$

(e) (f) (g) (h)

$H = (\Omega/2)\sum X + \sum V_{nn} r_{ij}^{-6} n_i n_j$
from $|+x\rangle$

R1: Ω/V_{nn}

$t/t_{\max}(g)$, $t_{\max} = 3/\Omega$

(i) (j) (k) (l)

$H = \sum r_{ij}^{-\kappa} Z_i Z_j + 0.1 \sum X$
from $|+x\rangle$

I1: κ

time t $[1/V_{nn}]$

(m) (n) (o) (p)

$H = (\Omega/2)\sum X + \sum \Omega R_b^6 r_{ij}^{-6} n_i n_j$
from $\bigotimes(\cos 0.1|g\rangle + \sin 0.1|r\rangle)$

R4: R_b/a

time t $[1/\Omega]$

(q) (r) (s) (t)

$H = \sum r_{ij}^{-3}(XX + YY + \Delta ZZ)/2$
$\Delta = J_z/J_{xy}$, from $|+x\rangle$

D3: J_z/J_{xy}

time t $[1/J_{nn}]$

(u) (v) (w) (x)

$H = -\sum r_{ij}^{-6} Z_i Z_j - h_f \sum X$
from gs$(h_f = 4)$

G4: h_f

time t $[1/J_{nn}]$

μ μ_{ph} $S_{N/2}$ [nats] $\log_{10} I$

t^*_μ t^*_{ph} t^*_{inf} χ_ϵ at cap experimental value